Research Statement

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Overview: Measuring Representation in Collective Decisions

Many collective decisions are now made with the help of algorithms: a voting rule decides which projects a city funds from its participatory budget, electoral systems turn votes into parliaments, and large language models are being tested for summarizing public consultations and drafting statements that a whole group can agree on. In each case we can ask: *does the outcome represent the people it affects?* Answering this with evidence requires a way to compare decision problems (are synthetic test cases similar to real ones?) and real data to test against. Computational social choice, the field that studies voting rules with the tools of computer science, long relied on proofs and on experiments with synthetic votes, often drawn uniformly at random, with no general method to check how close these were to real elections.

My work provides both, and takes them into practice:

- *Maps of elections.* I proposed the *positionwise distance*, which compares two elections by how often each candidate is ranked in each position, after optimally matching the candidates of the two elections, so it does not depend on candidate names or voter order. Drawing a collection of elections in the plane according to these distances gives a *map of elections*, on which similar elections lie close together and one can see which models of synthetic votes resemble real data [C3, J4].
- *Pabulib.* I initiated Pabulib, the largest open repository of participatory budgeting data. It holds over 2,000 elections from 8 countries and has been cited in more than 70 papers [C11].
- *The Method of Equal Shares in practice.* Under this rule, each resident receives an equal share of the budget, and a project is funded only if its supporters can jointly pay for it from their shares, so that each group of residents gets a fair part of the budget. I am a member of the team that convinced Polish cities to adopt it, and in 2025 I co-founded *Science for Democracy*, a nonprofit that works to bring the Method of Equal Shares to Polish municipalities.

As a PhD student, I was the principal investigator of a National Science Centre grant (NCN Preludium) on the *election isomorphism problem*, from which maps of elections grew. My thesis, *Map of Elections* (advisor: Piotr Faliszewski), received the Polish Prime Minister's Award (2025), and I received a START fellowship of the Foundation for Polish Science (2023) and the AAMAS 2023 Best Student Paper Award. As a CNRS postdoc at LAMSADE (Paris Dauphine-PSL) with Jérôme Lang (2024–2026), I worked at the intersection of social choice and machine learning. At the University of Geneva, I study collective decisions in international politics and in simulations with LLM agents [C28].

At EPFL, I will extend these tools to collective decisions in which opinions are expressed in text and aggregated by AI, such as summaries of public consultations, consensus statements, and AI agents that vote on people's behalf. For example, given the comments that residents submitted in a consultation and an AI-written summary of them, how can we measure which groups the summary leaves out?

Past and Current Research

How similar are two elections? A geometry of collective decisions. What an experiment on voting rules shows depends on which elections it contains: are the synthetic ones similar to real ones, and do they cover the situations a rule will face in practice? Answering this requires a distance between elections that ignores what the candidates are called and in which order the voters are listed, since two elections can have the same structure with entirely different

candidates (presidential candidates vs. types of sushi). We showed that the most natural such distances, and the related problem of finding one election inside another, are computationally hard [C2, J3, C5, J2]. The positionwise distance is my answer: it can be computed efficiently, and it made it possible to draw the *map of elections* [C3], to which we added landmark elections that make its regions interpretable [C4]; the complete framework appeared in *Artificial Intelligence* [J4].

The map turned a picture into a way of running experiments: it shows which parts of the space an experiment covers, which it misses, and where real elections lie relative to synthetic ones. The same geometry applies beyond ordinal elections: we built maps of approval elections [C7], of elections of different sizes [C19], of stable roommates and stable matching instances [C9, J1] (Best Student Paper Award), of tournaments [C22], and of fair division instances [C25], and used the geometry for preference learning [C8]. To help others use this methodology, I co-authored a guide to numerical experiments on elections [C15] and released two Python libraries: `mapel`, for drawing maps, and `prefsampling`, for sampling synthetic preferences.

Discovering structure in preferences. Beyond comparing elections, I develop measures of what is *inside* them. Polarization is usually measured in elections where each voter names one party. We introduced indices of agreement, diversity, and polarization for ranked ballots, built on *k-Kemeny*, a generalization of Kemeny's rule that seeks k representative rankings instead of one consensus, and extended them to approval ballots [C10, C27]. We then used them to quantify how diverse structured preference domains are [C24, C26]. A second line of work reverses the usual direction of the field: instead of designing a rule, we *discover hidden structure in a preference dataset*, such as consistent subelections [C13] or the most conflicting pair of candidates [C18], which brings social choice closer to machine learning. Closer to political science, we designed voting rules that let allied candidates avoid the spoiler effect [C14] and a nonparametric test for gerrymandering [C16].

Real data and real deployments. In participatory budgeting (PB), residents vote on how to spend part of a city's budget. I initiated Pabulib in 2020, and I built and maintain it [C11]; it is now the largest open library of PB data, used by researchers and by municipal policy teams. Real data opened questions that theory alone could not answer: learning realistic models of approval voting [C20], measuring how well individual projects performed under a given rule [C17], and a game-theoretic analysis of how proposers set project costs under the Method of Equal Shares [C23].

This work reaches practice: the team I belong to convinced Polish cities, including Wieliczka and Świecie, to adopt the Method of Equal Shares. This experience of bringing a voting rule from research into municipal practice shapes the tools I plan to build (Direction B).

Toward language. At CNRS LAMSADE, I designed simulation frameworks that model multi-agent decision-making. At the University of Geneva (Global Studies Institute), I study collective decisions in international politics, combining voting records with NLP pipelines that extract structured information from large collections of diplomatic and policy texts. Two projects show this direction.

Simulating democratic deliberation with LLM agents. With collaborators at the University of Toronto and Google DeepMind, we built *VoteSim*, which simulates a full democratic cycle with LLM agents [C28]. Voters, conditioned on the profiles and statements of real participants in the PRISM dataset, rank the parties' policies on twelve divisive issues; seven electoral systems turn the same ballots into seven parliaments; and each parliament negotiates a bill, which the voters then evaluate. Across all seven language models we tested, voters preferred the bills of proportional systems to those of majoritarian ones, and the least satisfied fifth of voters gained the most. Because the voters are simulated, these results describe language-model agents, not real electorates.

What an abstention means in the UN General Assembly. Abstention is the most common way states withhold support in the General Assembly (11.8% of roughly 860,000 votes in 1946–2025, against 5.5% votes against), yet standard measures of state preferences treat every abstention alike. In ongoing work, we classify all 101,070 abstentions by context (cast alone, inside a large crowd of abstainers, or against an organized opposing camp), using only the other states' votes and the abstaining state's voting bloc. Abstaining alone is concentrated among the permanent members of the Security Council, and the mix of contexts tracks international conflict while the overall abstention rate does not. We label the subject of each resolution by embedding its text with a language model and matching it to the UN library's thesaurus.

Research Agenda

Collective decisions increasingly involve text: free-text proposals, consultations with thousands of comments, and AI systems that summarize and mediate them. Social choice theory is not yet ready for this: its models treat the alternatives as abstract symbols, whereas a city project, a comment in an online consultation, or a UN resolution is a text with content. Recent work has made a promising start. *Generative social choice* (Fish et al., 2024) proves proportionality guarantees for slates of LLM-generated statements, assuming that the LLM answers its queries accurately, and the *Habermas Machine* (Tessler et al., 2024, in *Science*) showed that, in groups of five participants, statements written by an LLM mediator were accepted more often than those written by human mediators. In a position paper written with colleagues after a Dagstuhl seminar, we chart how text can enter every stage of collective decision making [P1]. What is still missing are ways to *measure and test* text-based outcomes, and *real data* on which to test them; my work provided both for ballots. My agenda has three connected research directions.

Direction A: Representation when opinions are expressed in text and aggregated by AI.

- **Auditing summaries and consensus statements.** Given opinions written in natural language and an aggregate text, such as a summary or a consensus statement, how should we measure how proportionally the aggregate represents the population? Which measures satisfy natural axioms, and which can be estimated with few LLM queries and provable error bounds? Existing methods *generate* representative text; we lack principled ways to *evaluate* it. Methods that find clusters and outliers among voters are a natural starting point for finding the groups that an aggregate text leaves out.
- **An LLM aggregator is a voting rule.** Fix a language model and a prompt that turns a set of opinions into k statements, and we obtain a multi-winner voting rule whose input is text. Its properties can then be tested like those of any other rule: does the outcome change when the same opinions are presented in a different order (anonymity)? Can a participant steer it by writing a manipulative opinion (manipulability)? Does it cover each group in proportion to its size? Large-scale numerical experiments of this kind are the methodology I helped establish for classical rules.
- **Polarization dynamics in AI societies.** When LLM agents deliberate and vote repeatedly, under which electoral systems and discussion structures do their positions converge, fragment, or polarize? Polarization indices for elections [C10, C27] provide the measures, and VoteSim [C28] provides the simulation environment.

Direction B: Participatory democracy as a living laboratory.

- **Votes over texts.** I will extend Pabulib with project descriptions, residents' comments, and outcomes over several years, turning it into a longitudinal benchmark in which every ballot refers to a text. The UN General Assembly record, with 5,693 resolutions adopted by recorded vote together with their full texts, is a second testbed of this kind. Both will serve Direction A as real data.
- **Does proportionality carry over to content?** Proportional rules for participatory budgeting have been observed to spread funds evenly across project categories (Nelissen, 2023; Peters

and Skowron, 2025). With project texts, we can ask whether they also give each group of residents a proportional share of the *topics* it cares about, and design rules that guarantee it.
- **Does the rule change the process?** We have analyzed how proposers choose project costs under the Method of Equal Shares [C23]. Field studies with partner cities can test how the rule affects real proposers and participants: who proposes, what they propose, and who votes.
- **Tools that cities and residents can trust.** Open tools to choose, simulate, and audit PB rules and to *explain* their outcomes, so that cities can adopt evidence-based designs and residents can see why a project was or was not funded.

I will seek partnerships with Swiss municipalities for the field studies, building on my experience of working with Polish cities through Science for Democracy.

Direction C: A geometry of collective decision problems, from ballots to text.

- **Computing invariant distances at scale.** Which domains admit polynomial-time relaxations of isomorphism-invariant distances that, like the positionwise distance, preserve the geometry? Can sketching and sublinear techniques make them scale to elections with millions of voters?
- **Benchmarks with provable coverage.** How many instances, and which ones, must a benchmark contain to cover the space of instances up to a given precision under a given distance? Which samplers achieve this? The answers will become defaults in `prefsampling` and `mapel`.
- **Maps of deliberation.** When alternatives are texts, two decision problems can be similar in how people voted, in what the alternatives say, or both. Distances that combine the two will make it possible to compare deliberations, human or AI, the way we now compare elections, and to place sessions of the UN General Assembly on a common map. This will give Direction A its evaluation backbone.

Goal for the first five years. By the end of this period, my group will have produced three things: a public benchmark of votes over texts, built from participatory budgets and the UN General Assembly record (B); open tools, with known guarantees, for auditing whether AI-written summaries and consensus statements represent the people they summarize (A); and maps that let researchers compare deliberations the way they now compare elections (C). Together, they would give the field for text what my earlier work gave it for ballots: shared data, shared measures, and a shared way of running experiments.

Students, funding, and service. The three directions offer PhD projects of different flavors: empirical and NLP-oriented (A), applied and data-centric (B), and theoretical (C). Natural first topics are testing LLM aggregators as voting rules (A), proportional coverage of topics in participatory budgeting (B), and scalable invariant distances (C). I have already co-supervised more than ten students, led a grant as principal investigator (NCN Preludium), and worked within an ERC-funded project. I will apply to the Swiss National Science Foundation early on and for an ERC Starting Grant built around Direction A. I serve on the editorial board of JAIR (since 2026) and review for NeurIPS, AAAI, and IJCAI.

Why EPFL

EPFL does not yet have a group in computational social choice, yet it is strong in the areas that this field now needs most: algorithms, natural language processing, and secure systems. I would bring the field to EPFL and connect it to these strengths. My agenda meets the work of three colleagues in particular:

- *Michael Kapralov*'s sublinear algorithms and sketching, for making isomorphism-invariant distances computable on elections with millions of voters (Direction C);
- *Antoine Bosselut*'s NLP group and EPFL's work on open multilingual language models such as Apertus, for auditing whether AI-written summaries represent people who write in different

languages, a natural question in a country with four national languages (Direction A);
- *Bryan Ford*'s work on proof of personhood and coercion-resistant e-voting, for digital participatory processes that count each resident once and protect the secrecy of the vote (Direction B).

Switzerland is also where I want to do this work. Few countries vote as often: Swiss citizens decide federal, cantonal, and municipal questions by popular vote several times a year, and in 2023 the city of Aarau used the Method of Equal Shares for its participatory budget. Geneva, where I already work, hosts the UN Human Rights Council, the natural next body for my work on voting in the United Nations.

Other references

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